

Project Charter

Project Identification	
Name or Title	Impact of Online and Offline Education on Student Performance and Engagement
Description	This project aims to analyze the comparative effects of online and offline education on student performance, engagement, satisfaction, and retention. Insights from the study will help inform strategic decisions on the future of educational delivery at St. Clair College.
Project Sponsor	Professor Pratik Bedi
Project Manager	Tran Bao Diem Phuong
Start Date	September 13 th , 2024
Finish Date	November 19 th , 2024
Estimated Budget	\$13,200

Project OBJECTIVES (purpose)
To compare the effect of online and offline education on student performance.
To evaluate the level of student engagement in online and offline education.
To identify the strengths and weaknesses of online and offline education.
To investigate factors influencing student preferences for online or offline education.
To recommend strategies for improving student outcomes in both online and offline environments.

PROJECT'S CRITERIA FOR SUCCESS (MUST BE MEASURABLE)
Student Performance Differences: Measure academic performance indicators like scores, grades, and completion rates to compare outcomes between online and offline education.
Engagement Insights: Track levels of student engagement through attendance, participation, and interaction rates with course materials and instructors in both settings.
Mode Preferences: Identify and quantify factors influencing student preferences for online or offline education, such as flexibility, interaction, or resource access.
Impact of Demographic Factors: Analyze how demographic aspects (age, location, socioeconomic status, etc.) affect student performance and engagement, providing measurable insights on variations across different student groups.
Challenges and Barriers: Document and quantify common challenges faced in each mode (e.g., technical difficulties in online education, time constraints in offline), assessing their impact on performance and retention.
Satisfaction and Feedback: Collect and analyze satisfaction scores or feedback ratings from students regarding each education mode to gauge perceived effectiveness.

Project SCOPE - HIGH LEVEL (INCLUDE PRODUCT AND PROJECT MANAGEMENT SCOPE)

Initiating

- ***Define Project Goals and Objectives:***

- Confirm the primary goals of comparing online and offline education impacts on student performance, engagement, and satisfaction.
- Finalize specific objectives, such as identifying strengths and weaknesses in each mode and recommending improvement strategies.

- ***Establish Project Team:***

- Identify and assign roles within the team, such as project manager, data analyst, survey designer, and report writer.
- Ensure each team member understands their responsibilities and project expectations.

- ***Secure Project Approvals:***

- Obtain approval from project sponsor (St. Clair College) and relevant stakeholders to proceed.
- Review and sign off on the initial project charter and scope with all parties.

- ***Preliminary Stakeholder Analysis:***

- Identify stakeholders, including students, educators, administrators, and policymakers who will benefit from the insights.
- Assess stakeholder needs and expectations to tailor the project outcomes and communication strategies.

Planning

- ***Develop Project Plan:***

- Create a detailed project timeline covering all major milestones, including survey design, data collection, analysis, and reporting.
- Assign resources, budget, and time allocations for each phase, ensuring efficiency within the 2-week timeframe.

- ***Risk Management Plan:***

- Identify potential risks (e.g., low survey response, data quality issues) and define mitigation strategies.
- Develop a risk register to monitor risks throughout the project lifecycle.

- ***Survey Design:***

- Develop the survey using Google Forms, focusing on questions that assess student demographics, engagement, performance, mode preferences, and challenges.
- Pilot the survey with a small sample to ensure clarity, relevance, and accuracy.

- ***Data Collection Strategy:***

- Define a strategy for distributing the survey to St. Clair College students through email and WhatsApp.
- Set response targets and plan reminders to ensure timely completion of data collection within the specified window.

- ***Data Analysis and Visualization Plan:***

- Plan for data cleaning, integration, and exploratory data analysis (EDA) to uncover patterns.
- Define data visualization requirements for clear presentation of findings using Tableau.

- ***Communication and Reporting Plan:***

- Outline a communication plan for sharing progress updates and final deliverables with stakeholders.
 - Schedule presentations and reports to keep stakeholders informed of project insights and outcomes.
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Executing

- ***Survey Distribution and Data Collection:***

- Launch the survey by sending emails and messages via WhatsApp to target participants, along with clear instructions.
- Monitor response rates in real-time, sending reminders to ensure a sufficient sample size within the data collection window.

- ***Data Cleaning and Integration:***

- Clean the data to handle any missing or inaccurate responses, document steps taken, and integrate it into a single dataset for analysis.
- Ensure data quality by addressing inconsistencies and outliers that could skew results.

- ***Data Analysis and Exploratory Data Analysis (EDA):***

- Perform EDA using statistical methods to identify trends, patterns, and key differences in engagement, performance, and preferences between online and offline settings.
- Use machine learning models if applicable, to predict outcomes based on demographic and mode-specific factors.

- ***Data Visualization:***

- Create interactive dashboards and infographics using Tableau, showcasing key insights on student performance, engagement, and satisfaction.
 - Ensure visualizations are accessible and understandable for both technical and non-technical stakeholders.
 - **Report Writing and Recommendations:**
 - Draft the final report, summarizing the findings and insights gained from the data analysis.
 - Develop a strategic recommendations document with actionable suggestions for enhancing online and offline education delivery.
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Monitoring and Controlling

- **Track Project Milestones:**
 - Use project management software or a tracking sheet to monitor each milestone's progress, ensuring adherence to the condensed timeline.
 - **Quality Assurance:**
 - Conduct quality checks at each stage (survey, data cleaning, analysis, visualization) to ensure accuracy and reliability of results. Review survey data quality, and validate data analysis methods and visualization outputs to maintain high standards.
 - **Risk Monitoring:**
 - Continuously assess and document any emerging risks, implementing pre-planned mitigation strategies as necessary.
 - Update the risk register with new information, and communicate any major risks or changes to stakeholders.
 - **Stakeholder Communication:**
 - Provide regular updates on project status, sharing intermediate findings if possible to keep stakeholders informed.
 - Address any concerns or questions from stakeholders promptly to maintain transparency and support.
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Project Closure

- **Finalize Project Deliverables:**
 - Complete and present all deliverables, including the final report, data visualizations, and recommendations document, ensuring they align with the project objectives.
 - Organize a formal presentation of the findings and recommendations for stakeholders, detailing insights and actionable strategies.
- **Gather Stakeholder Feedback:**
 - Collect feedback from stakeholders on project outcomes, recommendations, and overall satisfaction with the project process.
 - Use this feedback to evaluate the project's success and identify areas for improvement in future projects.
- **Document Lessons Learned:**
 - Conduct a project retrospective with the team to discuss successes, challenges, and lessons learned.

- Document any insights that could be applied to similar projects in the future to improve processes or strategies.

- ***Closeout Report:***

- Prepare a project closure report summarizing objectives, achievements, issues faced, resolutions, and any follow-up actions required.

- Include a summary of any unused budget and resources, and provide final financial reconciliation if needed.

- ***Formal Sign-off:***

- Obtain sign-off from the project sponsor and key stakeholders to officially close the project.

- Archive all project documentation, data, and reports in a secure location for future reference.

KEY PROJECT DELIVERABLES	
Name	Description
Survey Design and Distribution	A comprehensive survey designed to gather data on student demographics, education mode preferences, performance, engagement, and challenges. Distributed to target participants (e.g., St. Clair students) through platforms like email and WhatsApp, ensuring a broad and representative sample.
Data Collection and Cleaning Report	Raw dataset collected from survey responses, including demographic information and responses on engagement, performance, and preference factors. Data cleaning report documenting steps for handling missing values, correcting inconsistencies, and managing outliers.
Exploratory Data Analysis (EDA) Report	A detailed EDA report with visualizations and statistical summaries to identify patterns, trends, and initial insights regarding student performance and engagement in both online and offline settings.
Predictive Modeling and Analysis Results	Machine learning models to predict student outcomes based on survey data, such as engagement levels or academic performance across education modes.
Data Visualization Dashboards	Interactive dashboards created in Tableau to present key findings and comparisons, including engagement rates, performance metrics, student preferences, and identified challenges.
Final Research Report	A comprehensive report summarizing the study's findings, insights, and recommendations.
Recommendations Document	A strategic recommendations document detailing practical suggestions for educators and institutions to improve engagement, performance, and satisfaction in both online and offline modes.
Project Presentation	A slide presentation summarizing key project findings, insights, and recommendations for stakeholders such as educational institutions or policymakers.

Project Closure Report	A final report detailing project completion, objectives, successes, challenges, lessons learned, and any follow-up actions or future research opportunities.
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HIGH-LEVEL SCHEDULE		
Item	Major Events / Milestones	Dates
1.1	Project Initiation and Planning	September 13 th , 2024
1.2	Survey Design and Distribution	September 24 th - September 27 th , 2024
1.3	Data Collection Completion	October 10 th , 2024
1.4	Data Cleaning and Integration	October 11 th - October 13 th , 2024
1.5	Data Analysis and Visualization	October 14 th - October 25 th , 2024
1.6	Predictive Modeling and Reporting	October 26 th - November 5 th , 2024
1.6	Final Report and Stakeholder Presentation	November 6 th - November 19 th , 2024

CONSTRAINTS (Time, budget AND TECHNICAL)
Constraint Description
Time Constraints: - Project Timeline: Must be completed within a specific period covering survey distribution, data collection, analysis, and reporting. - Data Collection Period: Requires adequate time to gather a meaningful sample size. Potential delays in response rates could impact deadlines. - Analysis and Reporting Deadlines: Meeting these milestones is critical as data analysis and reporting are time-intensive.
Budget Constraints: - Survey and Data Collection Costs: May incur minor costs for survey distribution and participation incentives. - Software and Tools: Costs for data visualization and statistical tools like Tableau may apply, with open-source options as alternatives. - Human Resources: Limited budget may restrict team size, potentially leading to increased workloads and delays.
Technical Constraints:

• Data Quality and Accuracy: Ensuring reliable survey design for accurate data. • Technical Expertise: Project requires skills in data analysis, machine learning, and visualization. • Access to Tools: Requires tools like Python and Tableau. Licensing or access limitations may impact capabilities. • Privacy and Compliance: Must adhere to privacy regulations (e.g., FERPA, GDPR), which may limit data collection and analysis types.
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Assumptions
Description
Sufficient student participation and response accuracy.
Necessary tools (e.g., Google Forms, Tableau) and technical expertise will be available.
Data collected will be of high quality and relevant to objectives.
Sample will reflect a diverse student demographic. Sample will reflect a diverse student demographic
No significant changes in educational delivery during the project.
Compliance with privacy laws (FERPA, GDPR) is assumed.
Findings will yield actionable insights valuable to stakeholders.

RISKS	DESCRIPTION
Survey Participation	Sufficient Response Rate: There’s a risk of receiving a low response rate, which may affect the representativeness of the data. Mitigation involves sending follow-up reminders and possibly offering incentives to boost participation. Honest and Accurate Responses: There's a risk that participants may not provide honest answers, which could impact data accuracy. Mitigation involves clear instructions to encourage truthful responses.
Access to Resources and Tools	Availability of Analytical Tools: Necessary software and tools for data cleaning, analysis, and visualization (e.g., Python, Tableau) may not be readily available. Mitigation includes identifying suitable open-source alternatives or securing licenses in advance. Technical Expertise: There's a risk of limited access to team members skilled in data analysis and visualization. Mitigation may involve additional training or consultation with experts.
Data Quality and Relevance	Quality of Collected Data: Poor data quality could result from missing values or outliers. Mitigation involves a rigorous data-cleaning protocol to ensure data reliability. Relevance of Questions: If survey questions are not optimally designed, data might lack relevance to project objectives. Ensure questions are reviewed and piloted before distribution.

Representative Sample	Demographic Diversity: There's a risk that the sample may not be representative of the entire student body in terms of demographics, impacting generalizability. Education Mode Experience: If participants lack experience in one or both education modes, insights could be limited. Mitigation involves targeting students with known experience in both modes.
Stable Project Environment	No Major Changes in Education Delivery: Unplanned changes in the educational environment (e.g., a shift back to all in-person learning) may affect data relevance. Timeline and Budget Stability: Unforeseen delays or budget cuts could impact project scope or quality. Mitigation includes proactive contingency planning.
Data Privacy and Compliance	Participant Consent: Obtaining informed consent may be challenging or impact participation rates. Adherence to Privacy Laws: Compliance with data privacy laws (FERPA, GDPR) is critical to prevent legal issues. This involves careful planning of survey data collection and storage protocols.
Utility of Findings	Actionable Insights: Findings may not yield clear, actionable insights if data is too generalized or not properly interpreted. Stakeholder Interest: There's a risk that stakeholders may not find the results valuable or may be slow to adopt recommended strategies. Mitigation includes presenting findings in a compelling way and actively involving stakeholders throughout the project.
Machine Learning Model	Data Quality and Bias: Machine learning models rely on high-quality data to make accurate predictions. If the survey data contains bias (e.g., an unrepresentative sample) or errors, the model's results may be skewed. Model Selection and Accuracy: Choosing an inappropriate model type or hyperparameters could result in poor predictive accuracy, limiting the reliability of insights derived from the model. Overfitting: Overfitting occurs when a model learns too much from the training data, capturing noise instead of general patterns, which can lead to poor performance on new data.

SIGNOFF		
Signatures and Comments		
Name	Signature	Date
Tran Bao Diem Phuong	<i>Phuong Tran</i>	November 18 th , 2024
Devansh Sharma	<i>Devansh Sharma</i>	November 18 th , 2024
Gaurav Sunil Kumar Singh	<i>Gaurav Sunil Kumar Singh</i>	November 18 th , 2024
Ramandeep Singh	<i>Ramandeep Singh</i>	November 18 th , 2024
Keying Wang	<i>Keying Wang</i>	November 18 th , 2024
Comments:		

Key Stakeholder: *Prof.* Pratik Bedi

Project Team:

- Keying Wang
- Tran Bao Diem Phuong (*Project Manager*)
- Devansh Sharma
- Gaurav Sunil Kumar Singh
- Ramandeep Singh